# Safe Online Nonstochastic Control from Data

S. Kerz[1], A. Lederer[2], M. Leibold[1], and D. Wollherr[1]

[1]School of Computation, Information and Technology, Technical University of Munich
[2]Institute for Machine Learning, ETH Zürich

Contact: s.kerz@tum.de

## Motivation

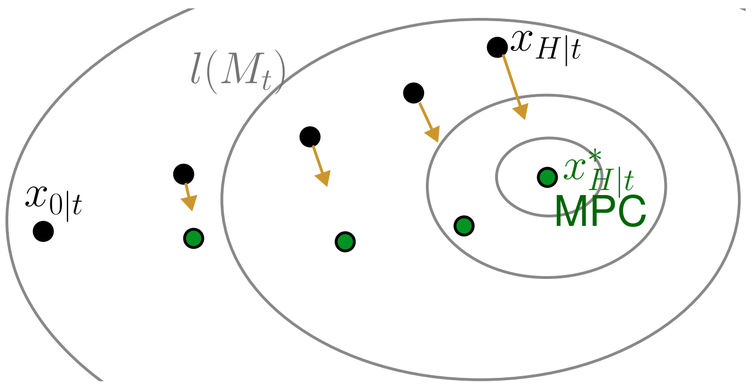
**Problem Setting:** Unknown linear system $x_{t+1} = Ax_t + Bu_t + w_t$ subject to nonstochastic disturbances $w_t \in \mathcal{W}$.

Online convex optimization (OGD) of Disturbance-Action Policies [1]: $u_t = Kx_t + \Sigma_{i=1}^{L} M_{i,t}\hat{w}_{t-i}$

Loss = cost of simulation $l(M_{t-1}) = c_t(x_{H|t}, u_{H|t})$.

**Goal:** Minimize time-varying costs $\Sigma\, c_t(x_t, u_t)$ while satisfying input and safety constraints $u_t \in \mathcal{U}$, $x_t \in \mathcal{S}$.

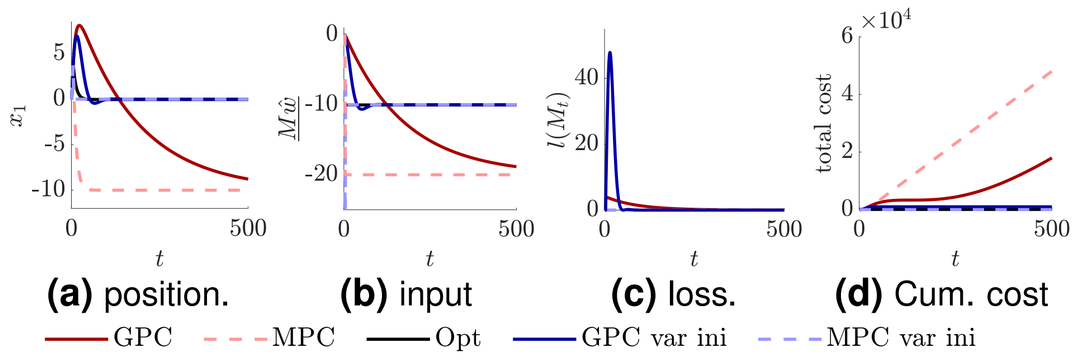
## Connection to Model Predictive Control




**Nonstochastic control chases MPC**

- predictions $x_{0|t} = x_t$ instead of hindsight cost $x_{0|t} = 0$
- optimal input sequence parameterized
- predictions include past disturbance estimates

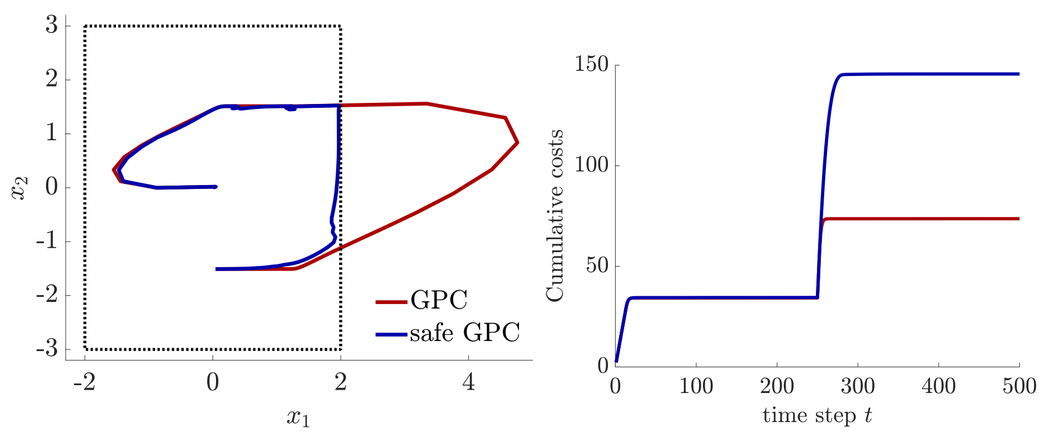
What if $x_{0|t} = x_t$ as in MPC? Convergence!




**(a)** position. **(b)** input **(c)** loss. **(d)** Cum. cost

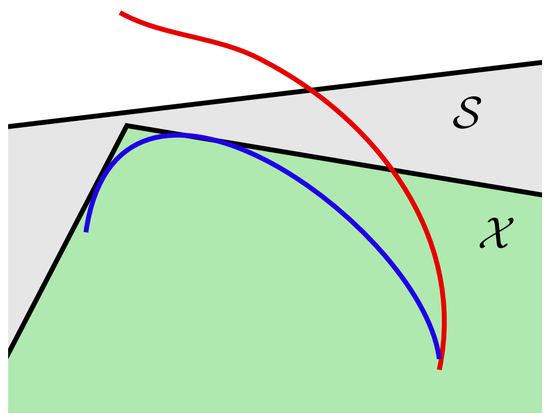
**Figure 1** Pathological example for constant disturbances and costs: GPC's loss tends to zero while the costs do not. With varying initial condition (var ini), the costs also tend to zero.

## Simulation Example




**(a)** State space **(b)** Cumulative costs

## Solution idea




**From data to a set of models** via set membership

$$\Omega = \{[A\ B] \mid x_{t+1} - [A\ B]\begin{bmatrix} x_t \\ u_t \end{bmatrix} \in \mathcal{W},\ t = 0, \dots, T-1\}.$$

Since $[A,\ B] \in \Omega \implies x_{t+1} \in \Omega \begin{bmatrix} x_t \\ u_t \end{bmatrix} \oplus \mathcal{W}$

**From a set of models to constraint satisfaction**

$$\Omega \begin{bmatrix} x_t \\ u_t \end{bmatrix} \in \mathcal{S} \ominus \mathcal{W} \implies x_{t+1} \in \mathcal{S}$$

**Recursive feasibility** Le $\mathcal{X}$ be robust control invariant

$$(\forall x \in \mathcal{X})(\exists u \in \mathcal{U})\ \Omega \begin{bmatrix} x \\ u \end{bmatrix} \in \mathcal{X} \ominus \mathcal{W}.$$

Plug in policy $\implies$ **set of safe parameters** $\mathcal{M}(\underline{\hat{w}}_t, x_t)$

**Including a $(\kappa, \gamma)$-strongly stable state feedback $K$**

$$\min_{c,Z,Y} c \quad \text{subject to } I_n \preceq Z \preceq cI_n,\ \begin{bmatrix} rZ & A_iZ + B_iY \\ * & rZ \end{bmatrix} \succ 0$$

then $K = YZ^{-1}$ guarantees $\|(A+BK)^t\| \leq \sqrt{c}\gamma^t$.

## Summary and Outlook

- Sharp safe safety without specific identification inputs
- OCO chases parameterized MPC, when to use what?
- Meaningful regret bounds missing

[1] N. Agarwal, B. Bullins, E. Hazan, S. Kakade, and K. Singh, "Online control with adversarial disturbances," in *36th International Conference on Machine Learning*, 2019.